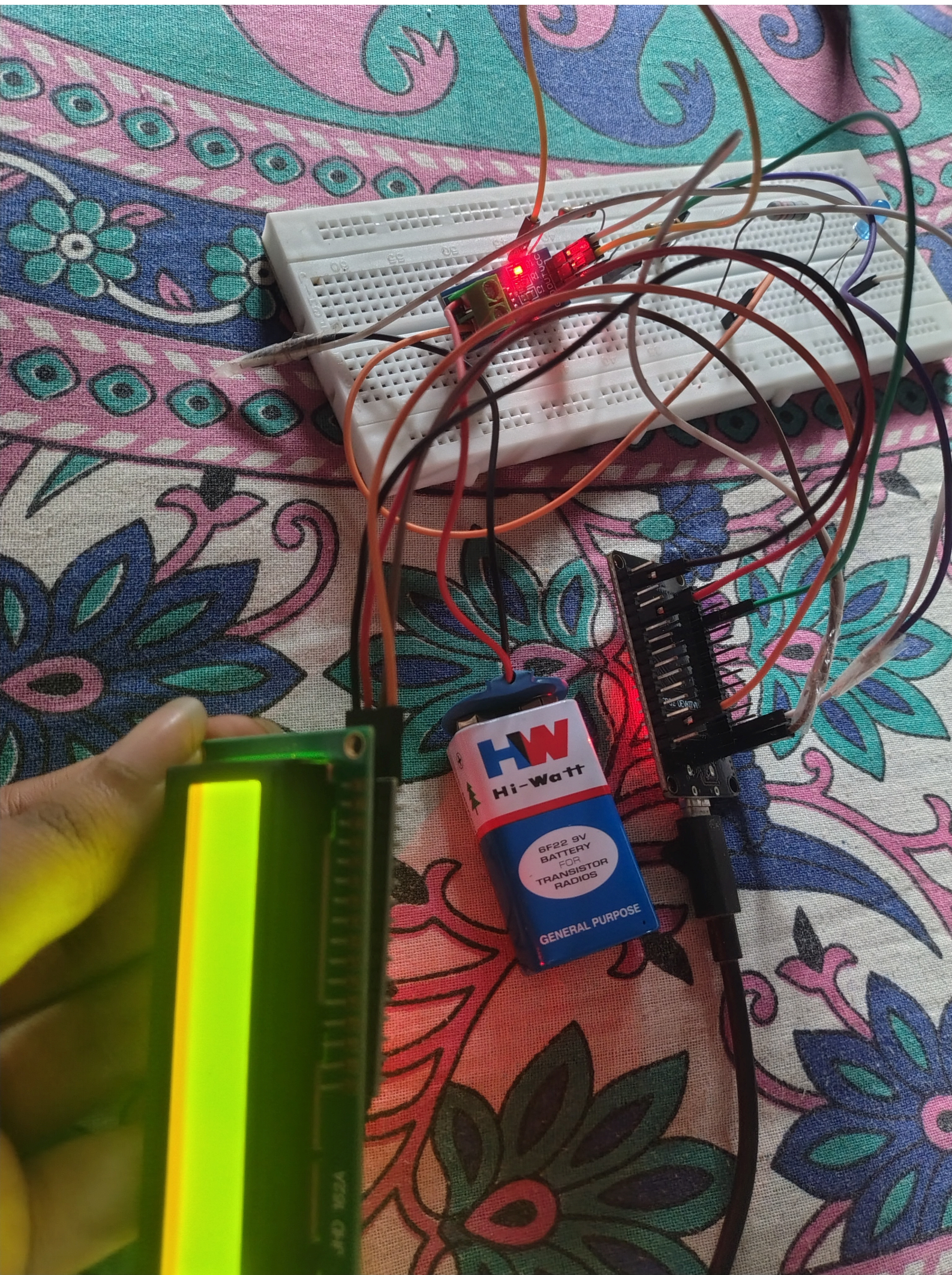
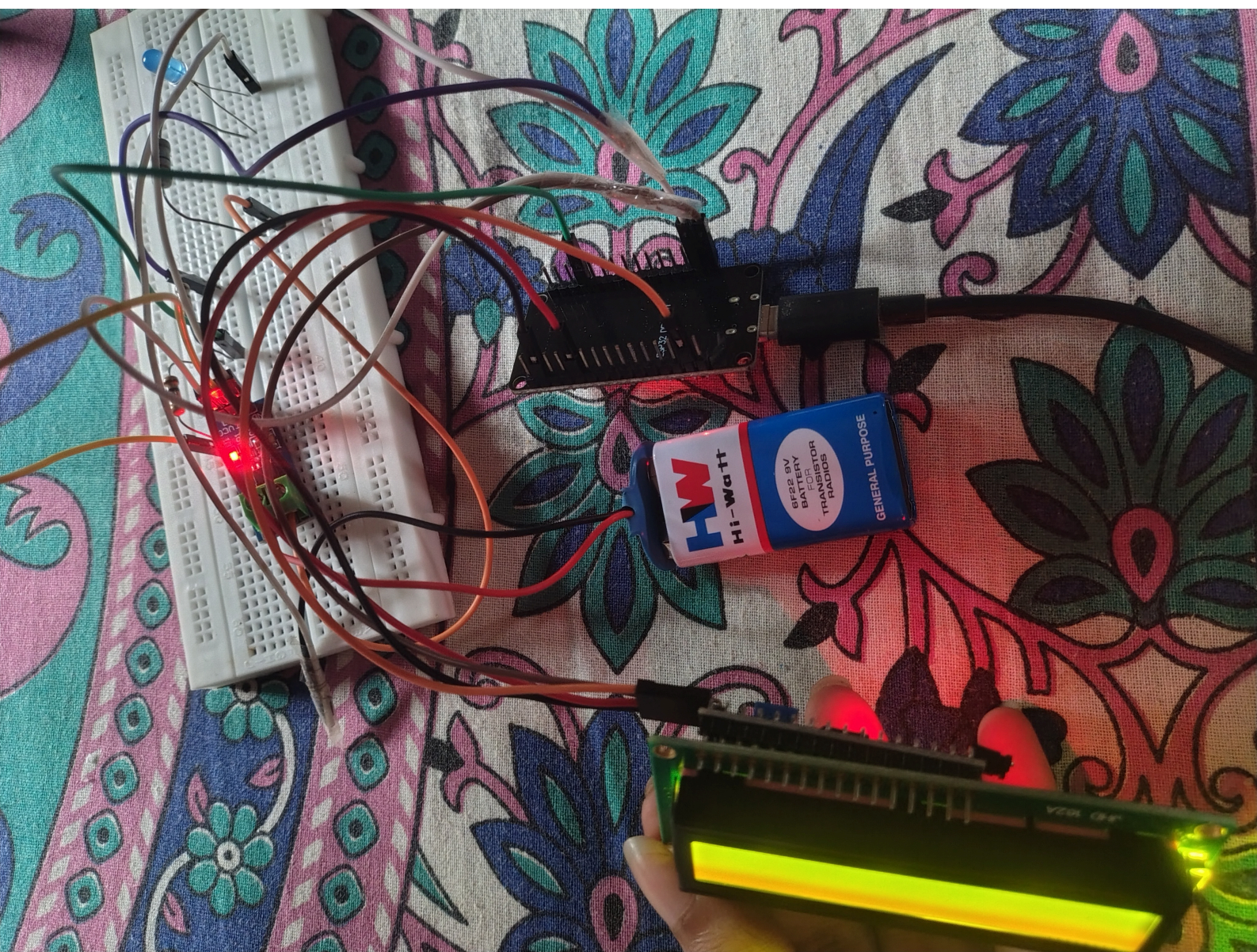


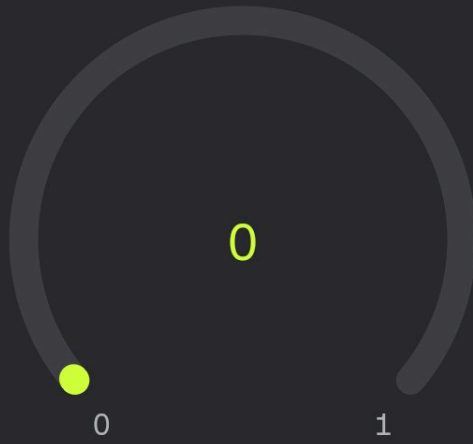




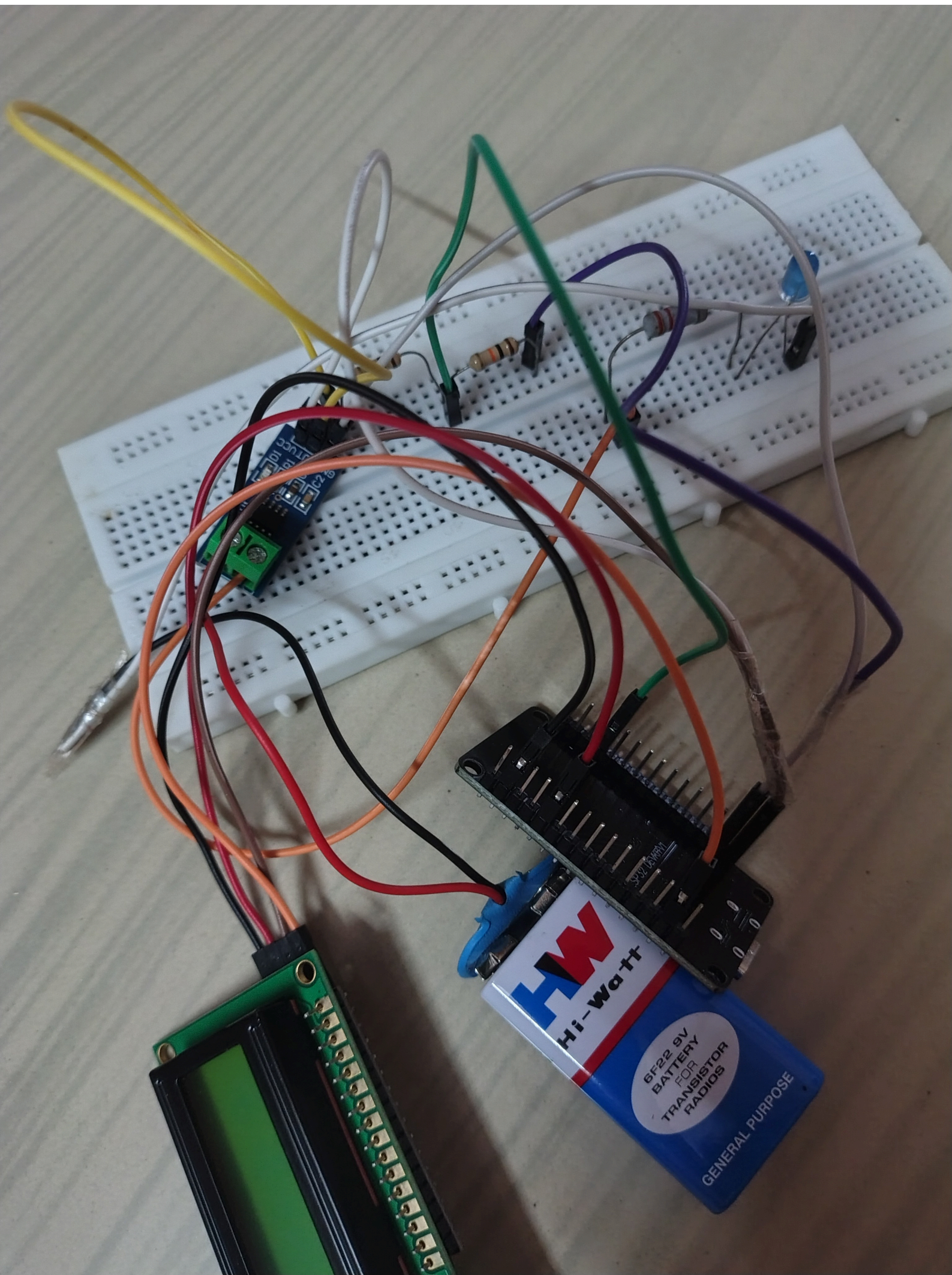




# IoT Smart Energy Meter ●









sketch\_sep5a | Arduino IDE 2.3.10

File Edit Sketch Tools Help

✓ → 🔊

ESP32 Dev Module

BOARDS MANAGER

Filter your search...

Type: All

Arduino AVR Boards by Arduino

1.8.8 installed

Boards included in this package: Arduino Duemilanove or Diecimila, Arduino Mini, Ardui...

More info

1.8.8 REMOVE

Arduino ESP32 Boards by Arduino

Boards included in this package: Arduino Nano ESP32

More info

2.0.18- INSTALL

Arduino Mbed OS Edge Boards by Arduino

Boards included in this package: Arduino Edge Control

More info

4.6.0 INSTALL

sketch\_sep5a.ino

```
1  #define BLYNK_TEMPLATE_ID "TMPL3TyZKSX41"
2  #define BLYNK_TEMPLATE_NAME "IoT Smart Energy Meter"
3  #define BLYNK_AUTH_TOKEN "3A0So9Mliul1hokIK8hvzeRdxza2uITnn"
4
5  #include <WiFi.h>
6  #include <BlynkSimpleEsp32.h>
7  #include <Wire.h>
8  #include <LiquidCrystal_I2C.h>
9
10 char ssid[] = "FAB.NET56";
11 char pass[] = "satabdi@123";
12
13 LiquidCrystal_I2C lcd(0x27, 16, 2);
14
15 const int ACS_PIN = 34;
16 const float sensitivity = 0.066; // ACS712 30A = 66mV/A
17
18 float zeroOffset = 0;
19
20 void setup() {
21   Serial.begin(115200);
22
23   lcd.init();
24   lcd.backlight();
25   lcd.setCursor(0,0);
26   lcd.print("Calibrating...");
27
28   pinMode(ACS_PIN, INPUT);
```

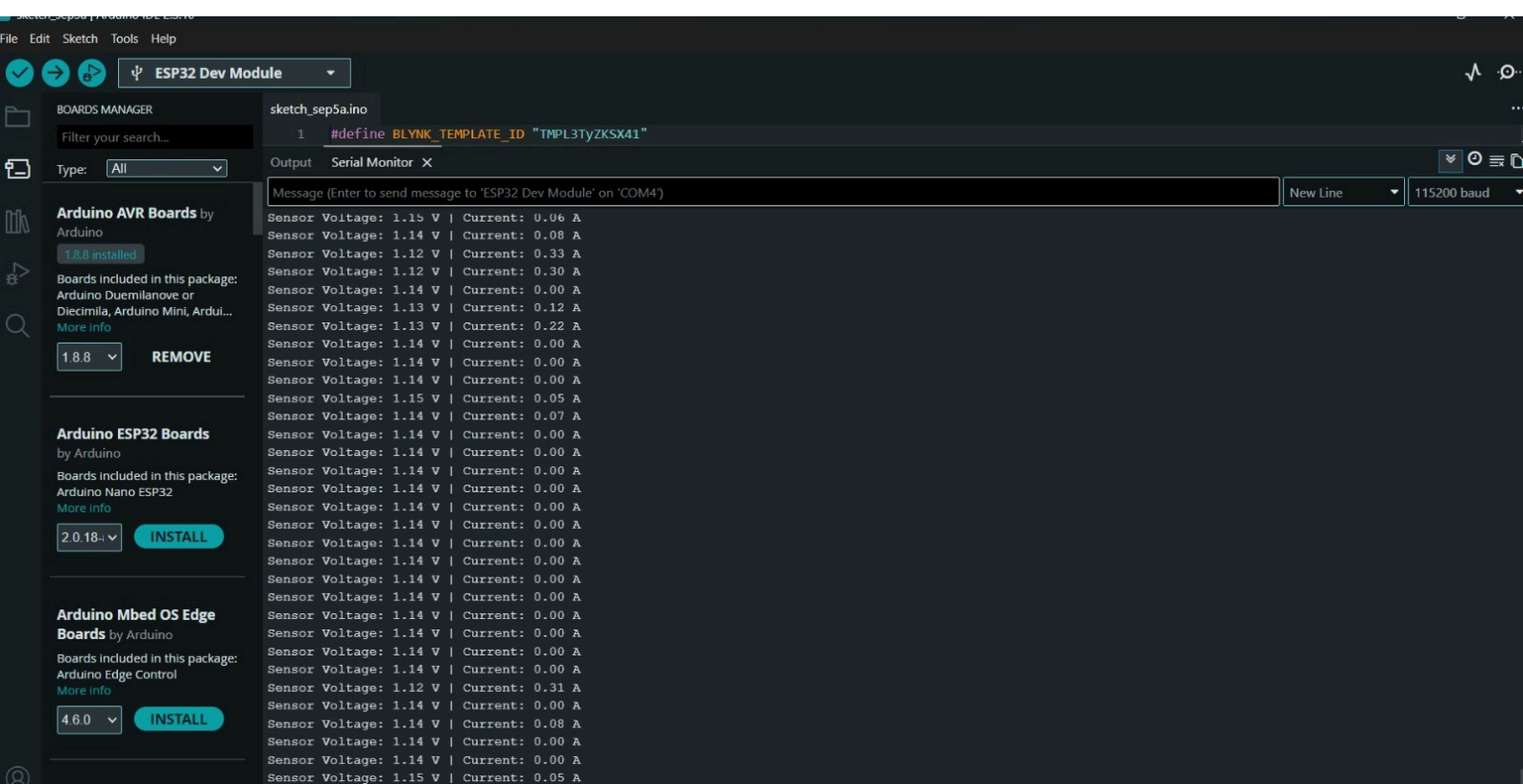
Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')

New Line 115200 baud

Sensor Voltage: 1.12 V | Current: 0.30 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.14 V | Current: 0.00 A







← ↻ 🔒 https://blynk.cloud/dashboard/750048/glob... ⚙️ ⭐️ ⚙️ ⭐️ Verify it's you Chat

**Blynk.Console** My organization - 6996PG Messages used: 305 of 100.0k Ask AI

Get Started Dashboards Custom Data Developer Zone

**IoT Smart Energy Meter** Online + Add Tag

Live 1h 6h 1d 1w 1mo 3mo 6mo 1y

voltage 8.85V current 0A Power 0W Total Energy 0

Cost 0

File Edit Sketch Tools Help

ESP32 Dev Module

BOARDS MANAGER esp32 Type: All

Arduino ESP32 Boards by Arduino Boards included in this package: Arduino Nano ESP32 More info 2.0.18- INSTALL

esp32 by Espressif Systems 3.3.11 installed Boards included in this package: M5Station, M5StampS3, Heltec Wireless Mini Shell, uPesy ESP32... More info 3.3.11 REMOVE

sketch\_sep5a.ino

```
1
2 #define BLYNK_TEMPLATE_ID "TMPL3Ty2KSX41"
3 #define BLYNK_TEMPLATE_NAME "IoT Smart Energy Meter"
4 #define BLYNK_AUTH_TOKEN "3A0S09Mliul1hokIK8hvzeRdxza2u..."
5
6 #include <Wifi.h>
7 #include <BlynkSimpleEsp32.h>
8 #include <Wire.h>
9 #include <LiquidCrystal_I2C.h>
10
11 char ssid[] = "FAB.NET5G";
12 char pass[] = "satabdi@123";
13
14 LiquidCrystal_I2C lcd(0x27,16,2);
15
16 const int ACS_PIN = 34;
17
18 void setup()
19 {
20   Serial.begin(115200);
```

Output

Writing at 0x000e6db6 [=====] 93.4%

Writing at 0x000ec43a [=====] 96.1%

Writing at 0x000f2301 [=====] 98.8%

Writing at 0x000f4f30 [=====] 100.0%

Wrote 937776 bytes (613698 compressed) at 0x00010000 in 11.3 s

Verifying written data...

Hash of data verified.

Hard resetting via RTS pin...



Search tabs

https://blynk.cloud/dashboard/750048/global/de... Verify it's you Chat

**Blynk.Console**

My organization - 6996PG Messages used: 9.6k of 100.0k Ask AI

**IoT Smart Energy Meter** Online + Add Tag

Live 1h 6h 1d 1w 1mo 3mo 6mo 1y H

**voltage** 1.15V

**current** 0.09A

**Power** 0W

**Total Energy** 0

**Cost** 0

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sketch\_sep5a Arduino IDE 2.3.10 File Edit Sketch Tools Help

ESP32 Dev Module

**BOARDS MANAGER**

Filter your search...

Type: All

**Arduino AVR Boards** by Arduino 1.8.8 installed

Boards included in this package: Arduino Duemilanove or Diecimila, Arduino Mini, Arduino...

1.8.8 REMOVE

**Arduino ESP32 Boards** by Arduino

Boards included in this package: Arduino Nano ESP32

2.0.18- INSTALL

**Arduino Mbed OS Edge Boards** by Arduino

Boards included in this package: Arduino Edge Control

4.6.0 INSTALL

sketch\_sep5a.ino

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2 #define BLYNK_TEMPLATE_NAME "IoT Smart Energy Meter"
3 #define BLYNK_AUTH_TOKEN "3A0S09Mliul1hokIK8hvzeRdxza2U..."
4
5 #include <Wifi.h>
6 #include <BlynkSimpleEsp32.h>
7 #include <Wire.h>
8 #include <LiquidCrystal_I2C.h>
9
10 char ssid[] = "FAB.NET5G";
11 char pass[] = "satabdi@123";
```

Output Serial Monitor X

Message (Enter to send message to 'ESP' New Line 115200 baud

Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.14 V | Current: 0.08 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.15 V | Current: 0.09 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.15 V | Current: 0.11 A  
Sensor Voltage: 1.13 V | Current: 0.14 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.12 V | Current: 0.30 A  
Sensor Voltage: 1.13 V | Current: 0.13 A  
Sensor Voltage: 1.13 V | Current: 0.23 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.15 V | Current: 0.05 A  
Sensor Voltage: 1.14 V | Current: 0.00 A  
Sensor Voltage: 1.15 V | Current: 0.09 A  
Sensor Voltage: 1.14 V | Current: 0.00 A



3

Blynk.Console

Get Started

Dashboards

Custom Data

Developer Zone

Devices

Automations

Users

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Locations

Snapshots

Fleet Management

In-App Messaging

Predictive Maintenance

My organization - 6996PG

Messages used: 9.6k of 100.0k

Ask AI

IoT Smart Energy Meter

Online

Auth Token: 3A0So9MI - Satabdi Paul My organization - 6996PG

🔔 ⚙️ 📄 ⚙️

Edit

Live 1h 6h 1d 1w 1mo 3mo 6mo 1y

voltage

1.13<sup>v</sup>

0 500

current

0.13<sup>A</sup>

0 100

Power

0<sup>W</sup>

0 1,000

Total Energy

0

0 1

Cost

0